Implementation Model for Access Control using Log Based Security

Practical Approach

Amritpal Singh
Department of Computer Science
Lovely Professional University
Phagwara, India
apsaggu@live.com

***Abstract*— This paper proposes an implementation for access control in SQL Server, and the technique is to improve security of the data masking technique devised for secure data warehouse since it stores the sensitive information which needs to be secure from attackers. A large number of techniques have been proposed for the same but based on analysis those techniques produce large overheads. This technique can be implemented in database management system of any kind. By the help of proposed technique, it will get to know the behavior of access strongly based on mining process on intelligently recorded log and other metadata of respective access.**

Keywords—Data Masking, DataWarehouse, Data Encryption, Intrusion Detection

I. INTRODUCTION

Data warehouse is an integrated repository derived from multiple source databases. With the quantity of data in corporation mounting day after day, many business enterprises are put together data warehouse to unify the accurate data flowing through their business into valuable storage area [1]. Protecting such data from attacks must be a chief priority for any enterprise security team. The objective of making massive amount of data accessible to users is an eminent one, but it is also one that necessitate rigid defense to keep data secure and obstruct illicit access [2]. Given the amount of data and the kind of data warehouse queries, the majority of the current data security explanation for databases is ineffective, using overly resources resulting overhead in query reaction time. Some security solutions are obtainable in relational database management systems such as Oracle 11g. There are mainly two types of data security answer available for data warehouse: preventive and reactive. Preventive security solutions are used to defend data in prior of attacks, such as encryption and data masking techniques for altering original data values. Data Masking is desirable solution for data security from both internal and external security intimidation. As the quantity of applications mounting, more and more information gets communal, thus escalating the threat of a information penetration, where precise data gets uncovered to illicit users. Oracle Data Masking handled this dilemma [3] by permanently interchanging the novel precise information with information that has similar kind and features as the novel precise information. Common methods of data masking include encryption, substitution and shuffling. Encryption is a superior form of data masking. Oracle has built up its Transparent Data Encryption in 10g and 11g edition of their database management system [4]. In the same way MySQL also provide data encryption functions. Even though regular encryption algorithms existing in major DBMS and are capable to offer strong information secrecy but encryption involves several costs such as additional retention area of encrypted information and operating cost in query answer time [5]. If the encryption is made uncomplicated for advance database functioning, secrecy will get exposed. Reactive security solutions involve detecting unauthorized access, auto retrieval, amongst others, which attempt shielding information from invaders able to get around protective security techniques. Intrusion Detection based on looking for difference from normal conduct. Numerous decision support queries have an informal character, where any portion of data can be retrieved. This indicates that it is very tricky to determine "usual user action" and "likely attack action", making it very difficult to make a distinction between usual and unusual action using existing intrusion detection systems. The question arises here how to improve the current intrusion detection systems efficacy without endanger data warehouse performance.

Growing volume of data warehouses hold private information, secrecy-protecting procedures will turn out to be more significant. Data privacy is major concern linked to data warehouse growth. The privacy of information is especially crucial, since in the present day business any information can be reclaim and damaged [6]. Hefty sum of information is continuously in risk if an illicit user can violate the security. Conversely accessibility of information to each person can trigger security risk to information. However protecting information is comparatively harder when diverse users fancy retrieving the information. So to address this problem data masking method have been projected based on modulus

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operator [7]. The reason for using modulus operator as most of the values in fact tables stored in warehouse are numeric values. It is inexpensive and uncomplicated to implement in any database management system, involving little hard work. If provoker discovers way around the security application and get straight admittance, provoker will only witness masked values in table stored in database. To make masking technique even more efficient we can take the benefit of history log of the security application (based on mathematical modulus operator) by which we get to discern whether it is usual access or attack, as a result it will help in intrusion detection.

II. ARCHITECTURE

Security application based on modulus operator which give back the remainder after division of one number by another .It act as intermediary between the users and masked database at the back end. To inquire the database the user application have to forward queries to the application, which mask them based on formula and store them in database which remains masked all the time .The benefit of having the masked database is that if attacker bypass the security application by any means, then he/she will see only the masked values over there. The security application make use of security keys with help of masking is performed. Only the authorized users will get the desired results. Basic architecture consists of:

i. User Application (Front End)
ii. Security Application (Middleware)
iii. Masked Database (Back End)

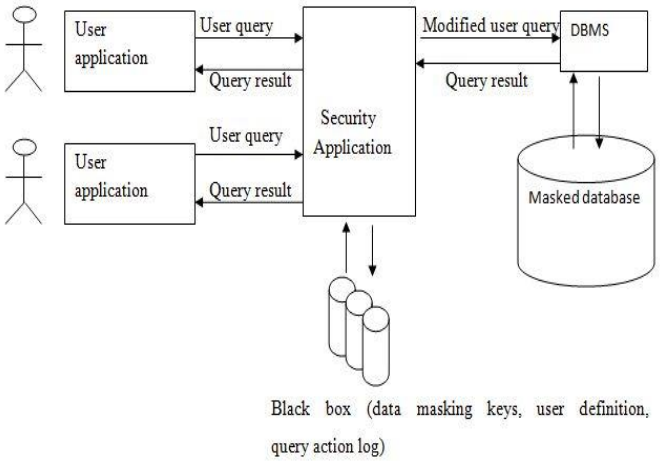

Fig 1. Basic Architecture

The data masking technique will work as follows: User application sends query to the security application which continuously monitoring and recording the user actions. Security application will mask the data input by user with help of key and store in database. Note that the data will store in database in masked form. Only the final and refined result will be sent back to authorized user. Every time user log in to application, log will be created which will be stored in black box which will contain the information regarding user identity, timestamp, operation performed by user whether he/she query the database with/without security application etc. In our solution we use 3 keys: two private keys and one public key. All keys are of 128 bit length.

III. DATA MASKING TECHNIQUE

Data Warehouse databases acquire large storage space .As we already know that data stored in form of numeric values in fact tables. In this section we will describe you the formula on basis of which masking and unmasking is performed. MOD is the modulus operator which finds the remainder after division of one number by another. MOD operator is non injective. For example we have table named customer with column credit card number need to be masked, we can mask with following SQL query:

ALTER TABLE customer ADD COLUMN K3

Here we are altering table customer by adding new column K3 which will act as public key. For each value of public key K3, we need to produce arbitrary value from 1 to 2^64. Keys K1 and K2 are confidential keys of 128 bit length stored in black box. The arbitrary values also need to be produced for K1 and K2 which are private keys from 1 to 2^64

UPDATE customer SET credit_num = credit_num - ((K3 MOD K1)MOD K2 + K2

For inquiry the database, we are familiar with the following SQL:

SELECT credit_num FROM customer

For unmasking, we need to perform following action. We require using instead the column creditcard_num with following SQL:

SELECT creditcard_num - ((K3 MOD K1) MOD K2) + K2 from customer

By unmasking, the user gets final and processed results. The Security application performs masking and unmasking automatically and store information about users in the form of logs every time the user performs any operation. Security application continuously monitors the situation and gives access only to authorized users which acts as middleware between users and database that receive the real query and replace each masked column name with relevant expression and send it to database management system to implement it which enhance privacy of system. This relates to whatsoever tables/columns have in the database.

IV. USING BLACK BOX

Black Box holds the user access course of actions and descriptions. Merely security application is able to get into it.

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Whenever user queries the database, query first submitted to security application, which redrafts the query and scrutinize the user approval in the Black Box. We have taken the advantage of Black Box in intrusion detection. In addition to access policies for users, it also contains private keys. It works similar to Wallet in Oracle which contains the encryption keys for database. As already explained, every time user queries the data, security application will generate log storing every single detail about user such as user name, what type of operation he/she has performed, timestamp etc. With help of logs stored in black box we have analyzed the behavior of user which helped us in differentiating the normal access and attack. The user credentials first match with information stored in black box, if it matches only then he/she granted access otherwise not. Black Box contains the analysis based register (Fig. 2) which stores the information about user obtain from logs [8]. Analysis Based Register consists of several fields like Log Sequence Number, Operation, Authorization, Transaction ID etc. Log Sequence Number (LSN) is exclusive id for a log record. User field tells us which user accessed the data.

LSN	00000014:00000061:0001
USER	U1
BEGIN TIME	22/02/2015 11:24:24 AM
TABLE	CUSTOMER
END TIME	22/02/2015 11:24:40 AM
OPERATION	INSERT
AUTHORIZATION	VALID
TRANSACTION ID	000:00000208

Fig 2. Analysis Based Register

Begin Time tells when user start performing any operation and End Time tells when he done with it. Operation field tells us what kind of operation he/she has performed such as update or insert. With help of authorization field we can find out whether user is legitimate or not. Transaction ID is inimitable for every action carried out by user. For example user U1 wants to query the masked database. First query is send to the security application which continuously monitoring the action of users. That action of user U1 is matched with data stored in the black box. If the security application finds it correct, then access is grant to user U1 and final result returned to user. Figure 3 shows generic log analysis flow.

V. Results and Discussions

At whatever time user accomplish a query, they present it to the security application, which redraft the query so as to process it with the original data values to substitute the particular masked columns used in the query, and scrutinize the user access approval in the Black Box. To redraft the user query, security application investigate for which tables and columns it desires to process, and search for the Black Box for reclaim K1 and K2 masking keys correspondingly, in addition to needed K3 key field in those tables.

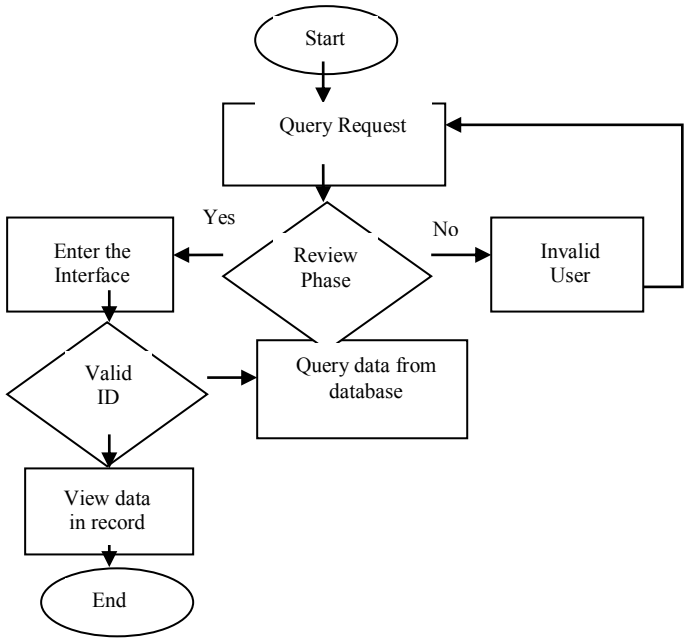

Fig 3. Log Flow

For example user inserted the data in table named employee with fields: id, salary and experience. Table 1 is given below.

TABLE I. EXAMPLE OF INITIAL DATASET

Id	Salary	Experience
20	10000	2
21	11000	3
22	12500	4
23	10500	1
24	13000	4

Table 2 shows the security application dataset. It is how the security application stores data in tables in database. The database remains masked all the times. If any user bypasses the application, then he/she will see only masked values. Table 2 is given below.

TABLE II. EXAMPLE OF SECURITY APPLICATION DATASET

Id	Salary	Experience	K3
20	2.3978E+46	1.6015E+46	5.60541E+46
21	3.0975E+46	2.3012E+46	4.90571E+46

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22	7.70781E+46	6.91151E+46	1.00052E+47
23	2.8981E+46	2.1018E+46	5.10511E+46
24	2.7981E+46	2.0018E+46	5.2051E+46

VI. CONCLUSIONS AND FUTURE WORK

We have taken the benefit of history log kept in black box for invasion finding. It can differentiate the normal access or attack based on behavioral analysis. It can implement in any database management system with an ease. As future work we can include automation for creation of our own data definitions. As security application works only for numerical values, in future we make sure application will able to mask alphanumeric values too.

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